Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/04/2025
PI Name:	Myers, Jerry Ph.D.		
Project Title:	Tempus ALS ISS Technology Demonstration (PI: Myers)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	216-433-2864
Organization Name:	NASA Glenn Research Center		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/08/2022	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Lemery, Jay	**Contact Phone:**	
Contact Email:	jay.lemery@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE: Dr. Beth Lewandowski, the previous Principal Investigator (PI), has retired and is no longer with NASA. Dr. Jerry Myer is the new PI for this investigation. Also, Kris Lehnhardt, M.D. and Benjamin Easter, M.D. are no longer with NASA and have been removed from the list of current Co-Investigators (Ed., 6/3/25).		
COI Name (Institution):	Suresh, Rahul M.D. (NASA Johnson Space Center) Thompson, Moriah (NASA Johnson Space Center) Schkurko, Courtney (NASA Glenn Research Center) Calaway, Kimesha (ZIN Technologies, Inc.) Miller, Rachael (KBR, Inc.)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The aim of this project is to evaluate the use of the Tempus Advanced Life Support (ALS) device in the spaceflight environment: • In collaboration with the European Space Agency (ESA), Phase 1 entails using the Tempus ALS device, compared to current International Space Station (ISS) medical devices, for the completion of Periodic Health Status exams and ISS medical contingency drills for 6 crewmembers (ESA = 2, NASA = 4), in order to achieve ESA's test objectives as described in the ESA Tempus ALS Technology Demonstration Activity Requirements Document (ARD)1. • NASA-led Phase 2 of the demonstration will entail Exploration Medical Capability (ExMC) and the Exploration Medical Integrated Product Team (XMIPT) working together to demonstrate the use of Tempus ALS on the ISS as part of progressively Earth-Independent Medical Operations (EIMO) through the execution of onboard medical simulations that emphasize crew autonomy during exploration medical care. The Tempus ALS is a multifunctional medical device with capabilities that include vital sign collection and monitoring, communications to enable telemedicine, interfaces with electronic health records, ultrasound, video laryngoscopy, and defibrillation. Tempus for ISS project activities will include evaluation of the usability of the Tempus ALS in a spaceflight environment, its functionality during the diagnosis and treatment of simulated medical conditions, and the increased onboard capabilities that it can provide when operated autonomously. The project results will be used to answer the question: Is the Tempus ALS a good candidate for inclusion within spaceflight medical system operations on exploration missions? For this project, simulations of one or more medical contingency scenarios will be executed and ISS astronauts will be required to follow and perform the NASA and ESA developed medical procedures necessary for diagnosing and treating the medical emergency. The astronauts will be required to use several of the capabilities of the Tempus ALS device while performing these procedures. The astronauts will perform the procedures under varying levels of autonomy, including with and without communication delays and with different types of procedural guidance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Tempus ALS device is designed for operations in pre-hospital and in remote settings. The Tempus ALS consists of two devices, the Tempus Pro (for vital sign measurements and clinical imaging), and the Tempus LS (for atrial defibrillation). The demonstrations of the Tempus Pro device performed during this study will provide additional user experience information as the device is used in the remote and extreme environments of the ISS and the Exploration Atmosphere Chamber. This information could further aid its effective use in other remote and extreme Earth-based environments.
Task Progress:	NOTE: Continuation of "Tempus ALS ISS Technology Demonstration" with new Principal Investigator (PI), Dr. Jerry Myer. The previous PI was Dr. Beth Lewandowski, who has retired from NASA. See the record under Dr. Lewandowski https:// for additional reporting (Ed., 6/3/25). During the fiscal year 2025, the final analysis, recommendation status of the Tempus Pro, and another Multi-functional Integrated Medical (MIM) device, the LifeBot, was developed from analysis of the data obtained from FY2024 testing by non-medically trained participants within the ground-based ExMC Technology Demonstration Test Bed, located at the NASA Glenn Research Center, the NASA Johnson Space Center (JSC) 20-ft hypobaric chamber, and the SpaceX Polaris Dawn Space flight. In each evaluation, the intent was to assess participants' ability to correctly utilize the device in different environments by collecting a NASA Task Load Index (TLX) survey after each activity and by recording the amount of time necessary to perform each activity. Final reports have been produced which provide more details and the key findings from the ground demonstrations. The remainder of fiscal year 2025 consisted of preparing the Tempus Pro for ground demonstration in the CHAPEA 1-year isolation project. This required Institutional Review Board (IRB) approval, the development of testing procedures commensurate with the CHAPEA environment, the preparation of the equipment and coordination of training material and training schedule with other CHAPEA participant studies. The participants in the CHAPEA study will evaluate the use of the Tempus pro to perform medical evaluations with training and video prompts and time-delayed feedback from ground support with simulated Mars delay. CHAPEA ingress is scheduled for late FY2025 with data availability in mid to late FY2026.
Bibliography Type:	Description: (Last Updated: 06/18/2025)
Conference Materials (Downloadable)	Haddix C, Lewandowski B, Miller R, Schkurko C, Nelson A. "Evaluating multi-functional integrated medical devices: insights from ground demonstrations." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 IWS2025_Haddix_MIMDeviceGS.pdf